

Strictly Singular Operators on L_p are $\mathcal{S}_1$ -Strictly Singular

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Status

This is a partial result adjacent to Mathieu and Tradacete's approximate factorization question in Remark 6.11 of [1]. It rules out the specific counterexample route proposed in the conditional packet `1810.09194_s1_schreier_obstruction_to_approx_factorization`, but it does not prove approximate (ℓ_r, ℓ_s) factorization.

Source Question

The source paper asks the following in Remark 6.11. The crop is from page 16 of the source PDF.

then X should contain a subspace isomorphic to ℓ_p , $\ell_{p'}$ or c_0 (although this last option is impossible because of the weak compactness of $L_A R_B$). Hence, the case $X = \ell_s$ as above can also be ruled out by this approach.

Remark 6.11. We do not know whether every $T \in \mathcal{S}(L_p)$ is approximately (ℓ_r, ℓ_s) -factorizable for some $r < s$. In fact, we do not know whether for an operator $T \in \mathcal{S}(L_p)$, and every $\varepsilon > 0$, there exist $r(\varepsilon) < s(\varepsilon)$, two sequences of finite dimensional spaces $(X_n)_{n \in \mathbb{N}}$, $(Y_n)_{n \in \mathbb{N}}$ and operators

$$\begin{aligned} T_1^\varepsilon &: L_p \longrightarrow (\oplus X_n)_{\ell_{r(\varepsilon)}} \\ T_2^\varepsilon &: (\oplus X_n)_{\ell_{r(\varepsilon)}} \longrightarrow (\oplus Y_n)_{\ell_{s(\varepsilon)}} \\ T_3^\varepsilon &: (\oplus Y_n)_{\ell_{s(\varepsilon)}} \longrightarrow L_p \end{aligned}$$

such that $\|T - T_3^\varepsilon T_2^\varepsilon T_1^\varepsilon\| \leq \varepsilon$. Keeping in mind the previous comment, by Remark 5.3 and the argument in the proof of Theorem 6.9, such a factorization would yield an alternative direct proof of [15, Theorem 2.9].

In connection with approximate factorization the following property of strictly

In text form, the source asks whether every $T \in \mathcal{S}(L_p)$ is approximately (ℓ_r, ℓ_s) -factorizable for some $r < s$, and asks a finite-dimensional-sum strengthening.

Definitions

The first Schreier family is

$$\mathcal{S}_1 = \{F \subset \mathbb{N} : F \text{ finite and } |F| \leq \min F\}.$$

An operator $T : X \rightarrow Y$ is $\mathcal{S}_1$ -strictly singular if for every normalized basic sequence (x_n) in X and every $\varepsilon > 0$ there exist $F \in \mathcal{S}_1$ and $0 \neq x \in [x_n]_{n \in F}$ such that

$$\|Tx\| \leq \varepsilon \|x\|.$$

Androulakis, Dodos, Sirotkin, and Troitsky introduced this hierarchy of Schreier strictly singular operators in [2].

Proof Intuition

The Kadec-Pelczynski dichotomy says that a seminormalized weakly null sequence in L_p has a subsequence modeled either on ℓ_p or on ℓ_2 . Thus, after passing to subsequences, both a candidate domain sequence and its image under a bounded operator have only two possible spreading geometries.

If a strictly singular T were not $\mathcal{S}_1$ -strictly singular, then on some basic sequence all Schreier-admissible blocks would stay bounded below. After the Kadec-Pelczynski reductions, the domain looks like ℓ_a and the image looks like ℓ_b , with $a, b \in \{p, 2\}$. If $a = b$, T is an isomorphism on the generated infinite-dimensional span, contradicting strict singularity. If $a \neq b$, boundedness forces $a < b$, and then long averages over $\{N + 1, \dots, 2N\} \in \mathcal{S}_1$ have image/domain ratio of order $N^{1/b-1/a} \rightarrow 0$.

Main Theorem

Theorem 1. *Let $1 < p < \infty$. Every strictly singular operator $T : L_p[0, 1] \rightarrow L_p[0, 1]$ is $\mathcal{S}_1$ -strictly singular.*

Proof. The case $p = 2$ is immediate because strictly singular operators on Hilbert space are compact, hence finitely strictly singular and therefore $\mathcal{S}_1$ -strictly singular. Assume $p \neq 2$.

Let (u_n) be a normalized basic sequence in $L_p[0, 1]$ and fix $\varepsilon > 0$. Since $L_p[0, 1]$ is reflexive, by passing to a subsequence we may assume that (u_n) is weakly convergent. If its weak limit is zero, set $v_n = u_n$. Otherwise pass to a further subsequence and set $v_n = u_{2n+1} - u_{2n}$. The sequence (v_n) is weakly null and seminormalized: normalized basic sequences are separated in norm. After normalizing and passing to a subsequence, the Kadec-Pelczynski dichotomy [3] gives an exponent $a \in \{p, 2\}$ such that

$$(x_n) := (v_n / \|v_n\|)$$

is equivalent to the unit vector basis of ℓ_a .

If a subsequence of (Tx_n) tends to zero in norm, then a singleton Schreier set gives the desired vector. Thus, after passing to a subsequence, assume that (Tx_n) is seminormalized. Since (x_n) is weakly null, so is (Tx_n) . Applying Kadec-Pelczynski again, pass to a further subsequence so that (Tx_n) is equivalent to the unit vector basis of ℓ_b for some $b \in \{p, 2\}$.

We claim first that $a \neq b$. If $a = b$, then for all finitely supported scalar families (α_n) the two equivalences give constants $c, C > 0$ with

$$\left\| T \sum \alpha_n x_n \right\| = \left\| \sum \alpha_n Tx_n \right\| \geq c \|(\alpha_n)\|_{\ell_a} \geq C^{-1} \left\| \sum \alpha_n x_n \right\|.$$

Thus T is bounded below on $[x_n]$, contradicting strict singularity.

Next, boundedness of T forces $a < b$. Indeed, the equivalences and boundedness imply

$$\|(\alpha_n)\|_{\ell_b} \lesssim \left\| \sum \alpha_n Tx_n \right\| \leq \|T\| \left\| \sum \alpha_n x_n \right\| \lesssim \|(\alpha_n)\|_{\ell_a}$$

for all finitely supported (α_n) , which is possible only when $a \leq b$. Since $a \neq b$, we have $a < b$.

Choose N so large that the constants in the two basis equivalences give

$$\frac{\left\| T \sum_{n=N+1}^{2N} x_n \right\|}{\left\| \sum_{n=N+1}^{2N} x_n \right\|} \leq \varepsilon.$$

This is possible because the numerator is $O(N^{1/b})$, the denominator is $\Omega(N^{1/a})$, and $1/b - 1/a < 0$. The set $F = \{N+1, \dots, 2N\}$ belongs to $\mathcal{S}_1$, so this gives the required Schreier small vector for the normalized weakly null sequence (x_n) .

It remains only to transfer this vector back to the original basic sequence (u_n) . If $v_n = u_n$, this is immediate, using that $\mathcal{S}_1$ is spreading under subsequences. If $v_n = u_{2n+1} - u_{2n}$, then a Schreier set $F \in \mathcal{S}_1$ in the difference sequence uses at most the original coordinates

$$\{2n, 2n+1 : n \in F\},$$

after the preceding subsequence selection. This set has cardinal at most $2|F|$ and minimum at least $2 \min F$, hence it is again in $\mathcal{S}_1$ after passing back to the original indices. Therefore the required vector lies in the span of an $\mathcal{S}_1$ -admissible subset of the original sequence. Since (u_n) and ε were arbitrary, T is $\mathcal{S}_1$ -strictly singular. $\square$

Corollary 1. *There is no operator*

$$T \in \mathcal{S}(L_p[0, 1]) \setminus \mathcal{S}_1(L_p[0, 1])$$

for $1 < p < \infty$.

Proof. This is exactly Theorem 1. $\square$

Relation to the Approximate-Factorization Problem

The preceding conditional packet proved that the finite-dimensional-sum factorizations in Remark 6.11, in the reflexive exponent range $1 < r < s < \infty$, force $\mathcal{S}_1$ -strict singularity. Corollary 1 shows that this necessary condition is automatic for strictly singular operators on $L_p[0, 1]$. Hence the Schreier-rank obstruction cannot yield a counterexample to Remark 6.11 on classical L_p .

The original approximate-factorization question remains open from this argument: $\mathcal{S}_1$ -strict singularity is only a necessary condition, not proved here to be sufficient for approximate (ℓ_r, ℓ_s) factorization.

Literature and Duplicate Checks

Cheap run-index searches for 1810.09194, `S_1`, `Schreier`, `strictly singular Lp`, and `Kadec` found the prior conditional packet and nearby Schreier-rank literature, but no existing packet proving this L_p endomorphism theorem.

Targeted web searches for phrases including “`S1 strictly singular Lp`”, “`Schreier strictly singular Lp`”, and “`not S1 strictly singular Lp`” found the Androulakis-Dodos-Sirotkin-Troitsky paper [2] but no direct statement of Theorem 1. The proof here combines the standard Schreier definition with the classical Kadec-Pelczynski subsequence dichotomy [3].

Human Review Recommendation

Review the transfer from arbitrary basic sequences to weakly null sequences or difference sequences, and verify the final Schreier-admissibility transfer for paired differences. The remaining proof is the standard two-exponent Kadec-Pelczynski dichotomy argument.

References

- [1] M. Mathieu and P. Tradacete, *Strictly singular multiplication operators on $\mathcal{L}(X)$* , arXiv:1810.09194, 2018.
- [2] G. Androulakis, P. Dodos, G. Sirotkin, and V. G. Troitsky, *Classes of strictly singular operators and their products*, arXiv:math/0609039, 2006.
- [3] M. I. Kadec and A. Pelczynski, *Bases, lacunary sequences and complemented subspaces in the spaces L_p* , Studia Math. 21 (1961/62), 161–176.